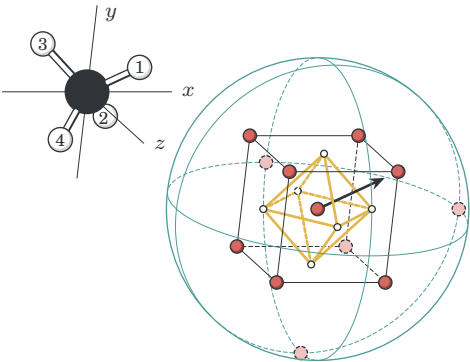
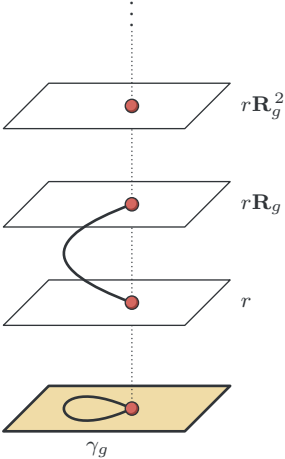


methane, CH_4 , the reference $\mathbf{X}_0$ in its body frame
 its twofold axes x, y, z pass through midpoints of opposite edges



the orientation space $SO(3)$ as a ball
 direction gives the axis and distance the angle
 the twelve rotations of T in red
 the cell $\mathcal{U}$ at $q = 0$ in gold
 the arrow lifts the third-turn about bond 1
 from r to $r\mathbf{R}_g$



over the base $SO(3)/T$ lie twelve sheets
 one for each rotation in T , three drawn
 the third-turn's path γ_g is a loop in the base
 lifted, it climbs from sheet r to sheet $r\mathbf{R}_g$
 A sees the climb as 1 , ${}^1\text{E}$ as $e^{2\pi i/3}$
 ${}^2\text{E}$ as $e^{4\pi i/3}$, and T as $x \rightarrow y \rightarrow z$